Frequent Losers: A Participation-Based Screen for Public Procurement

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Abstract

What signal can oversight bodies extract from contract-award records when bid microdata are unavailable? In São Paulo's electronic procurement platform (2009–2019), 2,735 of 16,843 firms with persistent zero wins participated in 13.5 or more tender-items; we call these firms *frequent losers*. Their presence associates with a $+3.6\% - +7.7\%$ higher negotiated unit price across estimators in the broad sample, but reverses sign under strict-overlap matching (-9.7% to -30.7%). The two estimates describe the same population at different scopes of selection. We report the broad-sample range as the headline because it maps to the deployment question—which environments to flag—and we treat the screen's signal as partly about selection rather than a uniform descriptive effect. The price imprint shrinks 12.6× across procuring-unit-size quartiles, consistent with thinner institutional capacity at smaller buyers, though we do not identify oversight as the operative channel. Against CADE's cartel records, the screen reaches AUC 0.748 on a strict contemporaneous benchmark and 0.864 on temporal holdout against cobidder labels—an indirect benchmark of cartel adjacency, not membership. The screen is a triage tool for procurement oversight where bid microdata are absent.

Keywords: frequent losers, participation-based screen, public procurement, cartel adjacency, triage, separating equilibrium

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1. Introduction

Brazilian electronic procurement has a peculiar feature: thousands of firms register, prepare bids, show up auction after auction, and never win. In São Paulo's Bolsa Eletrônica de Compras (BEC, 2009–2019), 16,843 firms bid without ever winning a tender-item, and 2,735 of them participated in more than 13.5 tender-items. We call these firms *frequent losers*.

Bid-rigging detection in the empirical literature relies on bid microdata that few jurisdictions audit at scale (Porter and Zona, 1993; Bajari and Ye, 2003; Imhof, 2019; Wallimann et al., 2023). Procurement systems instead universally preserve a much thinner envelope: contract-award records, with winner identity, participant identity, item identifier, and negotiated price. The question we ask is whether *this* envelope alone carries exploitable signal for triage. The answer is yes, but narrowly. Loser-side participation patterns associate with both higher negotiated prices and cartel-adjacent ground truth at rates above random matching, but the association is unstable across scopes and the validation labels are indirect. The participation-based screen we build prioritizes procurement environments for oversight; it does not adjudicate cartel membership.

Three findings, in descending order of empirical strength.. Within-item, within-year, within-procuring-unit comparisons across 1,654,401 tender-items associate frequent-loser presence with a $+3.6\% - +7.7\%$ higher log negotiated unit price across five estimators. Against CADE's cartel records, the screen discriminates 193 adjudicated cobidder firms inside the always-loser stratum at AUC 0.924 in-sample, 0.864 under temporal holdout, and 0.748 under a stricter contemporaneous benchmark restricted to four pre-2020 cases. Cobidders are firms that participate alongside adjudicated cartel members; the label is indirect by construction. Against the 47 direct CADE defendants in the broader BEC firm universe, the screen discriminates at AUC 0.49, by design: direct defendants are typically frequent winners and lie outside the target population. The price imprint shrinks from +21.4% in the smallest procuring-unit-size quartile to +1.7% in the largest—a 12.6×

ratio. We read the gradient as consistent with thinner institutional capacity at smaller buyers, but buyer size proxies for several correlated institutional features and we do not separately identify oversight as the operative channel.

The interpretive challenge.. The broad-sample price association reverses sign under strict-overlap matching: -9.7% overlap-cell ATT, -30.7% under propensity-score trimming. The broad-sample estimate combines a within-cell association with the systematic differences between items frequent losers select into and items they do not; strict-overlap matching isolates the former. Both scopes are descriptively valid. We report the broad-sample range as the headline because it maps to the deployment question (which environments to flag) and the screen targets the population frequent losers do select into, not a counterfactual matched on observables. A triage tool need not identify a stable causal effect; it needs to flag environments at higher prior risk than random selection, and on both the broad-sample association and the cartel-adjacency validation, the screen does so.

Theory.. We organize the construct through a separating-equilibrium framework with cover bidders and genuine entrants (Online Appendix A). The framework rationalizes the $\log(1 + \text{tenders_count})$ ranking and the wins $= 0$ filter, and it predicts a direction for the buyer-size gradient. It is an organizing device, not a source of identification.

What this paper does and does not do.. The participation-based screen flags procurement environments statistically associated with higher prices and cartel-adjacent validation targets. It is not a cartel detector, does not adjudicate firm-level guilt, and does not identify a stable causal effect of cover bidding on prices. The screen is designed for triage on award records and integrates with bid-distribution methods where bid microdata are available (§9.4).

The remainder of the paper proceeds in four blocks. Sections 2–5 situate the contribution, set out the institutional and legal framework, describe the BEC data, and state the descriptive empirical strategy alongside the design-based strategies that return null. Section 6 reports the CADE validation with explicit object discipline. Sections 7–8

present the within-item price association, the buyer-size heterogeneity, and the mechanism evidence. Section 9 collects the robustness work, and Sections 10–11 catalog scope limitations and conclude.

2. Related Literature

Four literatures bear on the screening problem. Each operates under a constraint on data or on identification timing: bid-coordination tests need the partition; proactive screens need bid microdata; cover-bidding identification has been retrospective; and the developing-economy literature has flagged the resource gap but has not closed it. The construct's positioning is at the conjunction of these constraints—not as a more powerful test against any of them individually, but as a screen that operates where the existing tests cannot, and that integrates with them where they apply (§9.4).

Bid-coordination tests require the partition as input. Bajari and Ye (2003) establish the canonical econometric framework for detecting collusion through exchangeability of bid functions and conditional independence of bid residuals; the tests achieve discriminating power against known cartels but require the researcher to first partition bidders into suspected colluders and a competitive fringe. The literature that builds on Bajari–Ye inherits this constraint: Porter and Zona (1993) and Porter and Zona (1999) detect bid rotation conditional on assumed cartel membership; Baldwin et al. (1997) document collusion in timber auctions identified through post-prosecution records; Conley and Decarolis (2016) infer bidder groups from co-bidding patterns—the closest precedent in data parsimony, but recovering group structure rather than individual flags; Clark et al. (2021) and Schurter (2020) study collusion dynamics in long-lived cartels with ex-post-known cartel membership. The partition is constructed *ex post* or by assumption in every case. The frequent-loser construct gives a candidate *ex ante* firm-level flag computable from contract-award records alone; bid-coordination tests can then be applied with the construct's flag as input where bid microdata are available.

Proactive bid-distribution screens require microdata that procurement systems do not always preserve. Imhof (2019), Imhof et al. (2018), Huber and Imhof (2019), and Wallimann

et al. (2023) train machine-learning classifiers on tender-level bid statistics (within-tender coefficient of variation, kurtosis, skewness, normalized spread); the most accurate implementations achieve AUC above 0.85 against ground-truth cartel cases in European procurement. Abrantes-Metz et al. (2006) pioneer the variance screen; Chassang and Ortner (2022) provide robust theoretical foundations for moment-based detection; Harrington (2008) surveys the field. Every implementation operates at the *tender* level and requires per-bidder bid amounts. Many real procurement systems do not preserve this layer at scale: electronic platforms, including BEC during much of our sample, archive winning prices and participant lists at the contract-record layer but not bid-by-bid. The gap is operational, not theoretical: what bid-distribution screens *can* do with full microdata is structurally larger than what oversight bodies *do* have. The frequent-loser construct runs on the second data layer and integrates with the first (§9.4).

Cover-bidding identification in the empirical literature has been retrospective. Porter and Zona (1999), Pesendorfer (2000), Asker (2010), and Marshall and Marx (2012) document cover bidders in prosecuted cartels through case records: the Ohio milk cartel, the Italian highway cartel, the New York City stamp cartel, and a sequence of cases reviewed by Marshall and Marx. Athey et al. (2011) compare cover-bidding incentives across auction formats theoretically. In every empirical case the cover bidders are identified through leniency testimony, post-conviction records, or judicial discovery—*ex post*, after the cartel is adjudicated. *Ex ante* prospective identification of cover-bidder candidates from public award-record data has not been attempted in the empirical procurement literature, to our knowledge. The construct inverts the retrospective workflow: by flagging firms whose participation patterns are coherent with cover bidding, the post-adjudication observation becomes a pre-investigation screen.

Procurement institutions shape both the supply of collusion and what screens can observe. A growing empirical literature documents how administrative discretion, repeated interaction, and political entrenchment alter procurement outcomes outside the cartel-detection lens. Coviello and Gagliarducci (2017) show that longer political tenure worsens procurement outcomes in Italy and is consistent with tighter official–bidder relationships;

Decarolis et al. (2020) show that discretionary procedures and limits to competition jointly predict corruption risk in Italian government contracting; Decarolis and Giuffrida (2017) documents the institutional channel through which procurement-officer characteristics moderate outcomes. These papers do not propose participation-only screens, but they motivate the procuring-unit-size oversight gradient we develop in §7.2: where oversight infrastructure thins, the participation footprint that the construct flags is amplified. The construct is most naturally read as identifying environments where institutional structure permits suspicious participation patterns, not as recovering cartel membership firm by firm.

Procurement enforcement is hardest where data are scarcest. The bid-rigging literature is dominated by developed-economy settings with mature data infrastructure, in which procurement spending represents 5–10% of GDP. In developing economies, where the share can exceed 12% of GDP (OECD, 2019) and oversight resources are thinner, settings are simultaneously more vulnerable to bid rigging and less equipped to deploy bid-distribution detectors. Sánchez Graells (2019) surveys the operational limits of existing screens in low-resource jurisdictions. The contribution generalizes in principle wherever a procurement system records winners and participants electronically, regardless of bid-by-bid archiving (§10.3).

Synthesis. The four literatures share a common structural constraint: each requires more information than typical oversight bodies routinely have, or operates only after collusion is established. Bid-coordination tests need the partition; bid-distribution screens need microdata; cover-bidding identification needs the post-adjudication record; and the developing-economy literature has flagged the resource gap without closing it. The frequent-loser construct addresses the conjunction. It produces a candidate flag that bid-coordination tests can take as input; it operates on the contract-award envelope that bid-distribution screens cannot use; it shifts cover-bidding identification from retrospective to prospective; and it is deployable in low-resource settings. The contribution is not a test that dominates any of the four benchmarks; it is a tool that runs where the existing tests cannot and that integrates with them where they apply.

3. Legal and Institutional Framework

Procurement statutes. Brazilian public procurement during our 2009–2019 sample is governed by Lei 8.666/93 (the General Procurement Statute, enacted to implement Article 37 of the 1988 Constitution) and Lei 10.520/2002 (which introduced the Pregão modality). Two modalities dominate BEC. *Convite* (Lei 8.666/93, Article 22) is a sealed-bid invitation procedure: the buyer invites at least three suppliers from the qualified-supplier registry, all proposals are submitted in writing within a five-business-day window, and the contract is awarded to the qualifying lowest bid. The statute requires three valid proposals for the procedure to clear; if fewer appear, the buyer must either re-issue the call or document the absence of bidders. *Pregão* (Lei 10.520/2002) is a reverse electronic auction: bidders submit progressively lower offers in real time until no firm improves the standing best offer. Pregão has no minimum-bidder requirement. Statutory value caps (Lei 8.666/93, Article 23) restrict convite to procurement below $R\$80{,}000$ for goods and services and $R\$150{,}000$ for engineering works during 2009–2017; Decreto 9.412/2018 raised these caps to $R\$176{,}000$ and $R\$330{,}000$ respectively. Above the caps, only pregão (or higher modalities not used in BEC) is permitted.

The structural difference between the two modalities provides useful institutional contrast for our analysis but is not the mechanism through which the construct operates. The convite minimum-bidder rule was introduced to ensure procedural competition; the older competition-economics literature (Marshall and Marx, 2012; Asker, 2010; Porter and Zona, 1993) has identified configurations of mandated bidder counts without corresponding rules ensuring genuine competition as canonical settings for regulatory cover bidding. The construct's empirical signal does not match this institutional reading: the price imprint and the discrimination of adjudicated CADE-cobidder firms are sharper in pregão (no minimum-bidder rule) than in convite (§7.1 and §8.2). We therefore present the convite statute as institutional context rather than as the channel the construct exploits.

The 2021 reform (Lei 14.133/2021) replaces Lei 8.666/93 for new contracts, restricts the convite modality, and tightens habilitation requirements (mandatory federal qualified-supplier registry, integrity programs above a value threshold). The reform postdates our

sample but shapes the policy relevance of the construct: by consolidating pregão-style electronic auctions as the institutional default, the reform moves Brazilian procurement toward the modal regime in which the construct's signal-to-noise is highest (§8.2, §10.3).

Enforcement architecture. Three institutions police BEC procurement. **Tribunal de Contas do Estado de São Paulo** (TCE-SP) audits purchasing units administratively, with binding power to require correction of irregularities, fine officials, and report findings to prosecutorial bodies; in 2023, TCE-SP processed over 12,000 administrative procurement cases at first instance. **Controladoria Geral do Estado** (CGE) oversees executive-branch compliance with procurement law and integrity programs, including the cadastros of debarred firms. The **Conselho Administrativo de Defesa Econômica** (CADE) is the federal competition authority; it investigates and adjudicates cartel and bid-rigging cases under Lei 12.529/2011, imposing administrative fines of 0.1%–20% of the infringing firm's gross revenue in the year preceding the investigation, and may refer cases to the Public Prosecutor's Office for criminal prosecution under Lei 8.137/90 (Article 3, prison sentences of two to five years for bid rigging). CADE's leniency program (Lei 12.529/2011, Article 86) grants immunity to the first cooperator and reduced fines to subsequent ones, generating the disclosure trail on which the cartel-detection literature relies for ground truth.

CADE's procurement-cartel portfolio relevant to our sample comprises 12 adjudicated cases covering 65 firm-defendants in total, of which 47 are active in BEC during the sample window. Final decisions span 2015–2025, with cases ranging from *Cartel dos trens-metrôs* (adjudicated 2019-07-08) to *Transporte escolar* (adjudicated 2025-02-26). Eight of the twelve cases were decided after our 2019 sample cut, so a portion of the consistency check in §6 operates prospectively: the construct's discrimination against the adjudicated cobidder population was computed on participation patterns observable years before enforcement confirmed those patterns. The interval between operational cartel activity and final CADE adjudication—the cases in our portfolio span 4–16 years from process opening to ruling—is the policy-relevant benchmark for the construct's contribution: a screen that flags suspicious environments in real time, even imperfectly,

compresses the case-load that reaches the forensic stage of the administrative pathway (§11).

The screening problem. Bid-level forensic tools—variance, kurtosis, skewness, and spread analyses across submitted bids—detect coordinated bidding when the required microdata are preserved (Imhof, 2019; Huber and Imhof, 2019; Wallimann et al., 2023). The information-theoretic logic of these tools is unimpeachable; their practical reach is bounded by data infrastructure. Many electronic procurement systems, including BEC during much of our sample, archive only winning prices and participant lists at the contract-record layer: bid-by-bid records may exist in operational logs but are not always exported into the analytical warehouses oversight bodies query.[2] The data asymmetry is not specific to Brazil. EU member-state platforms preserve bid microdata heterogeneously; the United States federal procurement data system (FPDS-NG) preserves contract awards but not bid history at the contracting-officer-accessible layer; subnational systems are typically thinner still. A screen that requires only participation records lowers the data threshold for proactive detection to the level virtually every system maintains. That is the construct's primary contribution to the law-and-economics of procurement enforcement: not a more powerful test, but a test deployable where the more powerful tests are not.

4. Data and Frequent Losers Definition

4.1. BEC Platform

The data come from the Bolsa Eletrônica de Compras (BEC), São Paulo's centralized electronic procurement platform, operated by the State Treasury since 2002. BEC serves 1,308 purchasing units (PBUs)—state agencies, public universities, hospitals, schools—and clears two procurement modalities during our sample: *convite* (sealed-bid invitation procedure with a statutory three-bidder requirement, Lei 8.666/93) and *pregão* (electronic reverse auction with no minimum-bidder requirement, Lei 10.520/2002). The

[2]In TCE-SP audits we examined, requesting bid-by-bid microdata from the BEC operational system is a multi-step administrative process that typically delays investigations by weeks or months; the contract-record layer queried directly produces results in minutes.

two modalities provide a within-platform institutional contrast that the empirical analysis exploits in §7.1 and §8.2; the construct itself does not depend on the convite minimum-bidder rule (§3). Our 2009–2019 window comprises 4,533,754 contract-record items across 22 calendar semesters, generated from 39,961,357 individual bids placed by 41,444 distinct firms (Table B.1). The contract-record layer preserves participants, winners, negotiated prices, reference prices, and procurement metadata; the bid-by-bid log is retained in the operational system but exported to the analytical warehouse only on administrative request, an envelope asymmetry central to the construct's portability claim (§10.3).

The modal mix shifts over the sample window. Pregão auctions expand from a minority share at the start of the sample toward near-universal coverage by 2019, consistent with the 2021 consolidation of pregão-style electronic auctions as the institutional default for Brazilian procurement (§3). Convite remains a meaningful share in the small-cap range and in early years, providing the modal variation that powers the falsification check of §7.1 and the discrimination comparison of §8.2.

4.2. Frequent Losers Definition

Behavioral anchor. Sustained participation with a zero win rate is rationally dominated under standard expected-profit competition. With positive per-bid cost $c > 0$, the expected gain per bid for a firm Bayesian-updating its win probability after n consecutive losses falls below c once $n > \bar{V}m/c - 2$, where $\bar{V}$ is the mean procurement-order value and m is the gross margin (Online Appendix A). At BEC parameters ($\bar{V}$ = R$86,000 from the sample mean, $m = 0.10$ from documented procurement-supplier margins), this threshold is approximately $n = 15$ for c = R$500 and $n = 41$ for c = R$200. The plausible cost range is anchored on documented BEC bid-preparation effort (one to two staff-hours plus documentation compliance, $c \geq$ R$200); at the FL threshold of 14 zero-win participations expected profit is already negative under that anchor, and at the 90th-percentile FL participation count of 50+ zero-win tenders cumulative direct cost exceeds R$10,000. Sustained participation past this point reveals one of three readings: a non-pecuniary objective (cover-bidding, capacity-signaling, or eligibility maintenance); systematically incorrect beliefs that fail to update on the loss record;

or zero-win participation in BEC offset by wins in unobserved tenders elsewhere. The construct flags the participation footprint these readings share; we adopt the cover-bidding reading as the working interpretation (§10.2).

Two-step classification. We define frequent losers as the intersection of two conditions.

1. **Always-losers**: firms with a zero win rate across all 2009–2019 BEC tenders. The pool comprises 16,843 firms (41% of all BEC participants), reflecting low electronic participation costs, small firms bidding beyond their operational capacity, and the broader compositional features of decentralized procurement (Coviello and Gagliarducci, 2017; Decarolis and Giuffrida, 2017). Frequent losers are a small extreme subset of this population.
2. **IQR threshold**: among always-losers, those with participation counts above median + 1.5 × IQR of the always-loser participation distribution.[3] The threshold yields 14 or more tenders with zero wins, corresponding to a statistical cutoff of 13.5. The threshold was fixed *ex ante* on the participation distribution and the behavioral anchor above, before any examination of CADE labels. The definition yields 2,735 frequent-loser firms.

The treatment variable losers_{igt} equals one if tender-item i in item group g at time t has at least one frequent-loser firm among its participants, and zero otherwise.

The economic construct is broader than the binary rule. The continuous loss-intensity statistic $\log(1+\text{tenders_count})$ is the primitive that Proposition 2 identifies as a sufficient ranking statistic for cover-bidder type given award-record data; the binary 14-tender rule is the information-coarsening of this statistic under the framework's separating equilibrium (Online Appendix A). The horse race in §8.1 establishes that the continuous statistic strictly dominates the binary in discrimination (DeLong $p = 2e - 05$). We treat

[3]We use median + 1.5× IQR rather than the standard Tukey rule (Q3 + 1.5× IQR) because the participation distribution is highly right-skewed; the rule we use yields a stricter threshold and fewer false positives. The referee's standard Tukey alternative flags 1,981 firms and yields AUC 0.834 ([0.804, 0.863]), versus 0.924 ([0.921, 0.926]) for the rule we adopt. Robustness to the multiplier choice across 1.0× to 3.0× is reported in §9.1.

the binary rule as the deployable operational implementation and the continuous statistic as the discriminating instrument.

Strict temporal validation. The classification pools participation over the full 2009–2019 window, which maximizes within-stratum discrimination but does not isolate the deployable projection. When we estimate the threshold only on the 2009–2016 training window, the cutoff falls to 7 and the firm-level AUC is 0.767 ([0.734, 0.800]). Frozen against 2017–2019 items, the binary indicator drops to 0.565, while the continuous loss-intensity statistic retains 0.770 (Table 2). The temporal-holdout audit in §9.3 reports the same gap on operational metrics: in-sample precision overstates the deployable projection by approximately 50% at top-500, and the temporal-holdout AUC of 0.864 is the conservative discriminating reference we adopt for operational claims. The binary rule is a deployable triage trigger; the continuous statistic is the more persistent identifier of the loser-side participation footprint.

Table 1: Loser-Side Concentration as Concept, FL as Operational Rule

Measure	Editorial role	Threshold	Full-sample AUC	Strict firm AUC	Strict item AUC
Continuous log(1 + tc)	Latent construct	—	0.939	0.750	0.770
Binary FL: median + 1.5 IQR	Operational rule	13.5	0.924	0.767	0.565
Alternative: Q3 + 1.5 IQR	Tukey alternative	18.5	0.834	—	—
Alternative: median + 1.0 IQR	Looser rule	10.0	0.902	—	—

Notes: The strict-train threshold estimated only on 2009–2016 is 7.0, far below the full-sample paper-rule threshold of 13.5. The revised paper attaches theory to the continuous loser-side concentration object and treats the binary FL rule as an administrative compression rather than a theoretically unique boundary.

Table 2: Strict Temporal Holdout with Threshold Frozen on 2009–2016

Scope	Score	Threshold	AUC	95% CI	Positives / N
Firm-level (AL)	Binary FL	7.0	0.767	$[0.734, 0.800]$	193/21,819
Firm-level (AL)	$\log(1 + \text{tc})$	—	0.750	$[0.706, 0.795]$	193/21,819
Item-level 17–19	Binary FL	7.0	0.565	$[0.564, 0.566]$	5,299/1,009,729
Item-level 17–19	$\log(1 + \text{tc})$	—	0.770	$[0.764, 0.776]$	5,299/1,009,729

Notes: Train-window threshold (2009–2016 AL): 7.0. Full-sample reference: 13.5. Binary FL threshold estimated only on 2009–2016; item-level rows apply the frozen threshold to 2017–2019 items.

4.3. Sample Construction

The participation distribution among always-losers is heavily right-skewed (Figures B.1–B.2). The regression sample retains convite and pregão tenders with a recorded winner and item types with at least one frequent-loser participation during the sample window. The last restriction conditions on the treatment in the sense that item types that never attract frequent losers anywhere in 2009–2019 are excluded; the restriction ensures controls come from item markets where the construct can plausibly fire. The unrestricted sample returns a coefficient within 0.5 percentage points of the restricted-sample baseline (§9), so the conditioning is empirically innocuous.

The resulting regression sample contains 1,654,401 tender-items across 18,783 item types, defined at the finest product-code level available in BEC (not aggregated to two-digit groups). Of the 18,783 item types, 15,101 (82%) have both frequent-loser-present and frequent-loser-absent tenders within sample, contributing to within-item identification under the fixed-effects specification of Equation (1). Frequent-loser firms participate in 4.8% of tender-items.

Table 3 reports unconditional descriptive statistics by frequent-loser presence. The unconditional negotiated-price ratio between frequent-loser-present and frequent-loser-absent tenders—R$140,673 versus R$12,465, a factor of approximately eleven—reflects the sorting of frequent losers toward larger contracts: frequent-loser-present tenders attract on average 9.06 bidders versus 4.90 in frequent-loser-absent tenders. Under within-item, within-year, within-procuring-unit fixed effects, the raw 2.43-log-point gap compresses to roughly 0.05 log points (Table B.2); the regressions compare similar tenders within tight product-code categories rather than across categories. Table 3 establishes the magnitude of the selection problem the fixed-effects structure is designed to absorb, and the comparison to Table B.2 shows the absorption is substantial.

4.4. CADE Validation Portfolio

The construct's external validation rests on the procurement-cartel adjudication portfolio of CADE, the federal competition authority (§3). The portfolio relevant to our sample comprises 12 adjudicated cases covering 65 firm-defendants in total, with final

Table 3: Descriptive Statistics: Tenders With vs. Without Frequent Losers

	With FL			Without FL		
	Mean	SD	N	Mean	SD	N
Negotiated price	140,673.37	3,148,893.84	79,452	12,465.30	484,167.26	1,574,949
Log negotiated price	4.92	3.87	79,452	2.49	2.59	1,574,949
Number of firms	9.06	6.35	79,456	4.90	3.17	1,574,991
Log number of firms	1.99	0.67	79,456	1.39	0.65	1,574,991
Number of bids	18.56	24.55	79,456	9.58	13.19	1,574,991
Non-FL firms	7.82	6.01	79,456	4.90	3.17	1,574,991
FL count per tender	1.25	0.75	79,456	0.00	0.00	1,574,991

Notes: Sample restricted to item types with $\geq$1 FL tender. FL = frequent losers defined by median + 1.5×IQR threshold.

decisions spanning 2015–2025. Of the 65 firm-defendants, 47 match BEC participants by CNPJ root, and these 47 firms constitute the *direct-defendant* ground-truth population. Eight of the twelve cases were adjudicated after 2019, so the construct's discrimination against the adjudicated population was computed on participation patterns observable years before enforcement confirmed those patterns—a prospective character we exploit in §6.

The validation operates against two distinct populations, and the distinction is central to the construct's claim. The *direct defendants* are the 47 firms named in CADE rulings as cartel members; in adjudicated cases these firms are typically frequent winners, the operational core of the collusive arrangement. The *cobidders* are the 193 firms that participated alongside the direct defendants in the same BEC tender-items, inside the always-loser stratum. Cobidders are cover-bidder *candidates* by structure: they share tenders with adjudicated cartel members without ever winning. The construct's target is the cobidder population, not the direct-defendant population. The AUC differential—firm-level 0.924 (95% CI [0.921, 0.926]) against cobidders versus 0.49 against direct defendants—is the empirical signature of this design (§6, §10.2).

5. Empirical Strategy

5.1. Estimand and Identifying Stance

The empirical object is β, the conditional mean difference in log negotiated price between tender-items that include at least one frequent-loser participant and otherwise comparable tender-items in the same item type, year, and procuring unit. We read β throughout as a *conditional association*, not as a causal effect. The descriptive stance is by design: a screen does not need to identify a causal parameter to allocate oversight case-load, and the operational pathway in §11 requires only that the construct flag environments at higher prior risk than random selection would.

Two design-based strategies that would extract a causal estimate from the data are reported and return null. A sharp regression discontinuity at the convite/pregão statutory value caps is gated by the absence of bunching: McCrary-style density tests around the $R\$80{,}000$ cap return a left/right ratio of 0.94 and a log-density discontinuity of -0.063 (Table B.5); the convite share drops cleanly across the cap from 69% below to 59% above, consistent with rule-driven assignment rather than strategic gaming. The local first-stage power is too weak to support a fuzzy RDD at standard bandwidths (Stock–Yogo F below threshold). A difference-in-differences design using Decreto 9.412/2018's 2018 raise of the convite cap fails the parallel-trends diagnostic at the available pre-treatment window. We report both failures openly and treat the empirical content as descriptive.

The descriptive claim is calibrated through four identification audits that operate alongside the reduced-form specification. The Cinelli and Hazlett (2020) robustness value $RV_{q=1} = 17.5\%$ and the Oster (2019) $\hat{\delta} = 261.64$ on the full sample bound the broad-sample association against selection on observables of the kind already absorbed by item, year, and procuring-unit fixed effects. Strict-overlap matching restricts the comparison to cells in which frequent-loser-present and -absent items genuinely overlap on observables, returning a sign reversal that we read as informative about selection within the always-loser stratum. A leakage audit decomposes the in-sample item-level AUC into a structural component (temporal-holdout AUC 0.864) and a pure-leakage component. A placebo using sub-threshold always-losers as instrument confirms that the

construct's price imprint operates through the above-threshold population we identify, not through generic always-loser supply. The four audits are reported in §9.2 and the identification scope they delineate is consolidated in §10.1.

5.2. Reduced-Form Specification

The reduced-form specification is

$$y_{igt} = \beta \cdot \text{losers}_{igt} + \mathbf{x}'_{igt}\boldsymbol{\delta} + \alpha_g + \lambda_t + \gamma_k + \varepsilon_{igt} \tag{1}$$

where y_{igt} is the outcome for tender-item i in item group g at time t and procuring unit k; losers_{igt} indicates the presence of at least one frequent-loser participant; $\mathbf{x}_{igt}$ includes a convite indicator and any specification-specific controls; α_g, λ_t, and γ_k are item, year, and procuring-unit fixed effects; and ε_{igt} is the disturbance, clustered at the item level. The headline outcome is $\log p_{\text{negotiated}}$; auxiliary outcomes (number of bidders, number of bids, number of non-frequent-loser bidders) are reported in §7.1 and the appendix.

We estimate Equation (1) under four specifications. The general within-item-and-year baseline drops the procuring-unit fixed effect; the within-procuring-unit specification adds it. Two modal subsamples, pregão-only and convite-only, decompose the headline along the institutional contrast set up in §3. The estimation sample comprises 1,654,401 tender-items (§4.3).

Three complementary estimators bracket Equation (1). Cross-fit estimation defines the frequent-loser list on odd years, estimates Equation (1) on even years, and reverses the assignment, recovering the structural component of β free of within-sample classification noise. Matching estimators adjust for selection on observables: coarsened exact matching (CEM) on year, procedure type, item group, and procuring-unit-size quartile, and inverse-probability weighting (IPW) on the propensity of frequent-loser presence given the same observables. A leave-one-out instrumental-variable specification uses the supply of frequent losers active in other procuring units within the same product market and year as instrument; we read $\hat{\beta}_{\text{IV}}$ as a measurement-error diagnostic rather than as a primary estimator, since the binary frequent-loser indicator carries classical attenuation toward

zero when it proxies for the underlying continuous loss-intensity statistic (§8.1). The four estimators (cross-fit, IPW, OLS, CEM) jointly bound the headline range $+3.6\% - +7.7\%$ that we adopt as the reference throughout the paper (§7.1).

Standard errors are clustered at the item level as the baseline choice. Robustness to procuring-unit clustering and to two-way Cameron et al. (2011) clustering is reported in §9.1; the point estimate is mechanical to clustering choice, and significance is preserved under all three regimes.

5.3. Heterogeneity Specifications

We characterize heterogeneity in β along two dimensions that match distinct framework predictions. Both are descriptive specifications of Equation (1) estimated within subsamples; neither is institutional identification.

The first dimension is procuring-unit size, which the framework of Online Appendix A maps to the principal cost of detection θ_k through the comparative static $\partial m^*/\partial\theta_k < 0$. We split the sample into quartiles of procuring-unit size, defined by annual contract volume in BEC, and re-estimate Equation (1) within each quartile separately. Robustness to alternative size measures (cumulative item count, headcount of distinct purchasers) is reported alongside the headline result. The specification anchors the procuring-unit-size oversight gradient of §7.2.

The second dimension is the cell-heterogeneity grid in winner concentration and repeated-pair density. For each frequent-loser firm j, we compute the Herfindahl–Hirschman Index of winning firms across all tenders in which j participates, $\text{HHI}_j = \sum_w s_{wj}^2$, where s_{wj} is the share of j's tenders won by firm w. We classify j as high-HHI if HHI_j exceeds the sample median and low-HHI otherwise; orthogonally, we classify j as high-pairs if j shares at least two tenders with at least one repeat partner among the winners, and low-pairs otherwise. The 2×2 cross-classification yields four cells, and we estimate Equation (1) with cell interactions to characterize whether the conditional price gap varies across cells. The specification anchors the cell-heterogeneity discussion of §8.3; we adopt the cell interpretation as descriptive scope, not as a positive test of cartel-rotation predictions.

6. Validation Against CADE Adjudications

The validation asks whether the construct discriminates populations of firms with a higher prior probability of cartel adjacency than random selection. It does not speak to firm-level membership. The labels we use are indirect by construction.

6.1. Validation populations

Table 4: Validation populations and object discipline

Population	Size	Definition and role
All BEC participants	41,444	Universe of firms bidding in BEC, 2009–2019.
Always-losers	16,843	Firms with zero wins across the sample. The construct discriminates within this stratum.
Frequent losers	2,735	Always-losers above the median + 1.5×IQR participation threshold. The construct's flagged population.
CADE direct defendants in BEC	47	Named in CADE rulings; typically frequent winners; not the construct's target.
Cobidders inside the always-loser stratum	193	Always-losers that participated alongside CADE direct defendants. The label of cartel *adjacency*.

Notes: The construct discriminates within the always-loser stratum, not across the broader BEC universe. Cobidders are firms that participated alongside adjudicated cartel members; participation alongside is consistent with cover bidding, capacity-transfer motives, market intelligence, or bystander participation in cartel-affected markets. Strong AUC against this population is not equivalent to identifying membership.

6.2. Conservative benchmark

The conservative benchmark restricts the ground truth to four CADE cases adjudicated before 2020 (coleta de lixo, aquecedores solares, trens-metrôs, tecnologia-informação 2019-10), covering 30 firm-defendants, all matching BEC. Always-losers co-bidding with at least one number 210, of which 108 are frequent losers. The participation-stratified co-bidding rate is 3.95% versus 1.24% at the participation-quartile-stratified baseline ($p < 0.001$, 1,000 iterations), an excess ratio of 3.2×. The ROC AUC against the 210-positive contemporaneous ground truth is 0.748 (95% DeLong CI $[0.713, 0.783]$).

This is the benchmark we treat as primary. The construct uses no information from cases adjudicated post-sample, and the labels come from cartels whose existence and

membership were established by enforcement before the construct was estimated. Within the always-loser stratum, frequent losers co-occur with adjudicated cartel members at rates above what random matching at comparable participation volume would produce.

6.3. Prospective benchmark and within-firm enrichment

A more permissive benchmark uses CADE's full procurement-cartel portfolio: 12 adjudicated cases, 65 firm-defendants, of which 47 are active in BEC. Among 2,735 frequent losers, 193 (7.1%) co-participate with at least one of the 47; the participation-stratified excess ratio is 3.5× ($p < 0.001$). The firm-level AUC is 0.924 (95% CI [0.921, 0.926]) in-sample and 0.864 ($[0.859, 0.870]$) under temporal holdout (train 2009–2016, test 2017–2019). This benchmark treats post-sample adjudications as informative labels for cartel involvement during the BEC window—a reasonable but not innocuous assumption. We report it as complementary, not as the headline.

A within-firm exercise on the same portfolio: 3 of 7 always-loser direct defendants (43%) cross the frequent-loser threshold against a population baseline of 16%. Table 5 lists them.

Table 5: Direct CADE defendants flagged by the construct

Firm	Cartel	CADE process	Tenders	Wins
Sol Tecnologia	Solar water heaters	08012.001273/2010-24	84	0
Nova Esperança Locadora	School transportation	08700.005876/2019-85	97	0
Jofran Comércio	Trash bags (Op. Colludium)	08700.005789/2015-02	65	0

These three firms are direct defendants who happen to share the always-loser footprint. They are the exception within the direct-defendant population, where most firms are frequent winners; the construct's AUC against direct defendants in the broader BEC firm universe remains 0.49 by construction.

6.4. Robustness to the enforcement record

Dropping all 31,447 CADE-involved tender-items and re-estimating the within-PBU baseline yields $\hat{\beta} = 0.062$ ($N = 1{,}622{,}954$), virtually unchanged from 6.4%. The price imprint operates independently of the enforcement record. Either the residual sample

contains undetected cartels the construct picks up before enforcement does, or frequent-loser presence correlates with higher-price environments through channels orthogonal to documented collusion. Both readings are consistent with the triage interpretation.

6.5. What the validation does and does not show

Three pieces of evidence anchor the triage interpretation: a 3.2× contemporaneous excess at AUC 0.748, a 3.5× prospective excess at AUC 0.924 (0.864 holdout), and the price-gap insensitivity to the enforcement record. The construct prioritizes procurement environments connected to adjudicated collusion at rates well above random matching at comparable participation volume. It does not detect cartels: against the 47 direct defendants in the broader BEC universe the AUC is 0.49, which places a hard upper bound on broader detection claims outside the always-loser stratum.

7. Main Results

This section reports two findings. The first is a positive within-item price association in the broad sample that reverses sign under strict-overlap matching; we treat the reversal as central rather than as a robustness check. The second is a steep heterogeneity in procuring-unit size, consistent with thinner institutional capacity at smaller buyers but not identifying oversight as the operative channel.

7.1. Within-Item Conditional Price Gap and Selection Scope

Table 6 reports OLS estimates of Equation (1). Frequent-loser presence associates with a higher log negotiated unit price in all four specifications: 6.8% within item and year (6.4% adding procuring-unit fixed effects), and 9.3% in pregão versus 3.8% in convite. The comparison set is the 1,654,401 tender-items in the same product-code × year × procuring-unit cells that do not contain frequent losers.

The convite minimum-bidder rule (Lei 8.666/93, Art. 22, § 3) does not generate the gap by forcing quorum-fillers: the imprint is sharper in pregão, where no minimum-bidder rule applies (+9.59% vs. +3.92%; Table B.4). Cross-fit estimation yields 3.6%, coarsened exact matching +0.077, and inverse-probability weighting 0.055. A leave-one-out IV

Table 6: Negotiated Prices (log): Effect of Frequent Losers

	(1) General	(2) General	(3) Pregão	(4) Convite
FL presence	0.0677***	0.0636***	0.0933***	0.0382**
	(0.0230)	(0.0215)	(0.0255)	(0.0186)
Convite	-0.0090**	0.0095***		
	(0.0037)	(0.0035)		
Observations	1,654,401	1,654,401	546,549	1,107,852
R-squared	0.8790	0.8860	0.8821	0.8932
Item FE	YES	YES	YES	YES
Year FE	YES	YES	YES	YES
PBU FE	NO	YES	YES	YES

Notes: Standard errors clustered at the item level in parentheses. *** $p<0.01$, ** $p<0.05$, * $p<0.1$. Columns (3) and (4) restrict the sample to pregão and convite procedures.

returns 0.194 ($p < 0.05$, first-stage $F = 396$), in the direction expected under classical attenuation when the binary indicator proxies for an underlying continuous participation count. The conditional range across these four broad-sample estimators is $+3.6\% - +7.7\%$.

The sign reversal under strict-overlap matching.. Restricted to cells in which frequent-loser-present and -absent items genuinely overlap on observables, the conditional coefficient *reverses sign*: overlap-cell ATT -9.7%, propensity-score-trimmed ATT -30.7% (Table 9). The broad-sample estimate combines two things: the within-cell association where the two populations overlap, and the systematic differences between items frequent losers select into and items they do not. Strict-overlap matching isolates the first; the broad-sample estimate carries both. Within overlap cells, the price comparison runs the opposite way, which likely reflects the unobserved features that drive entry into the selected cells. Cinelli and Hazlett (2020)'s $RV_{q=1} = 17.5\%$ and Oster (2019)'s $\hat{\delta} = 261.64$ bound the broad-sample association against selection on observables, but neither addresses the selection mechanism the strict-overlap exercise surfaces.

We report the broad-sample $+3.6\% - +7.7\%$ as the headline because it maps to the deployment question—which environments to flag—and a triage tool sees the population frequent losers do select into, not a counterfactual matched on observables. The within-cell coefficient answers a different question and is reported in parallel, not in opposition. We do not claim the broad-sample range as a uniform descriptive effect on prices; we claim that frequent-loser presence flags environments in which observable selection patterns correlate with higher prices and with cartel-adjacent ground truth.

7.2. Heterogeneity by Procuring-Unit Size

Splitting the sample into quartiles of procuring-unit size and re-estimating the within-item specification produces a steep, monotonic gradient: $+21.4\%$ in the smallest quartile, $+9.8\%$, $+4.5\%$, and $+1.7\%$ in the largest, all significant at conventional levels with item-clustered standard errors. The $12.6\times$ ratio is the largest source of heterogeneity in the data, larger than the pregão–convite gap, and is preserved across alternative measures of buyer size (annual contract volume, cumulative item count, headcount of distinct purchasers; Table B.3).

The framework in Online Appendix A predicts a gradient in the same direction: $\partial m^*/\partial\theta_k < 0$, where θ_k indexes the procuring unit's principal cost of detection. The mapping is suggestive, not unique. Buyer size proxies for several correlated institutional features, including procurement-officer tenure (Coviello and Gagliarducci, 2017), internal-audit infrastructure, item composition, and discretionary procedure use (Decarolis et al., 2020; Decarolis and Giuffrida, 2017). We present the gradient as consistent with thinner institutional capacity at smaller buyers, not as evidence that oversight is the operative channel. Identifying that channel would require variation in oversight plausibly orthogonal to the other correlated features—a research design we do not have. The policy implication (oversight bodies operating on small-buyer environments face thinner ex-ante information) is robust to the mechanism interpretation; the mechanism interpretation is not.

8. Mechanism Heterogeneity

The price gap and the procuring-unit-size gradient establish that the construct has a measurable, institutionally patterned imprint. This section reads the heterogeneity through the framework's mechanism predictions. Three subsections take three interpretive stakes: §8.1 shows that the continuous loss-intensity signal dominates the binary cutoff in discrimination, consistent with the framework's identification of $\log(1 + \text{tenders_count})$ as the sufficient ranking statistic; §8.2 reads the modal asymmetry between pregão and convite as scope information about where the construct fires most cleanly; §8.3 catalogs the mechanism predictions that the data do not support and demotes them to scope.

8.1. Loss Intensity Dominates the Threshold Cutoff

The deployable binary rule that defines frequent losers (more than 13.5 tender-items, zero wins) is an operationalization, not a primitive. Proposition 2 identifies $\log(1 + \text{tenders_count})$ as a sufficient ranking statistic for cover-bidder type given award-record data; the binary filter is the information-coarsening of this latent posterior under the separating equilibrium of Online Appendix A. The horse race in Table 7 confirms the prediction empirically.

Table 7: Horse Race: FL14 Binary vs Continuous Loss Intensity (Same Sample)

Specification	Coef on FL14	Coef on log(1+ max tc)	Convite	N
(1) FL14 only	+0.0653***	–	+0.171***	1,653,658
	(0.0216)		(0.013)	
(2) Continuous only	–	+0.0188***	+0.169***	1,653,658
		(0.0055)	(0.013)	
(3) Both, additive	−0.0746*	+0.0349***	+0.169***	1,653,658
	(0.0383)	(0.0108)	(0.013)	
Firm-level AUC against 193 *CADE-cobidder labels (always-loser pool,* $N = 16{,}843$*)*				
FL14 binary	AUC = 0.911 [0.898, 0.925]			
log(1+tenders_count)	AUC = 0.939 [0.932, 0.946]			
DeLong test (continuous vs FL14)	$Z = -4.30$, $p = 1.7 \times 10^{-5}$			

Notes: Outcome is $\log p_{\text{negotiated}}$ within the v13 panel of 1,653,658 items. Item-level treatment is $\mathbf{1}$[any FL14 participant] in column (1) and the log of the maximum loss-intensity score among always-loser participants in column (2). Column (3) includes both. The FL14 coefficient flips sign when continuous intensity is added to the specification: this is the disciplinary diagnostic that the binary discretization at 14 tenders is an operationalization of the underlying continuous signal, not an economically primitive threshold. We report the binary specification as a deployable implementation rule and the continuous specification as the substantive object. SEs clustered at item level. DeLong tests use the firm-level AUC reported in the bottom panel. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

On a harmonized same-sample basis, the continuous $\log(1 + \text{tenders_count})$ score discriminates the 193 adjudicated CADE-cobidder firms inside the always-loser stratum at AUC 0.939 (95% CI [0.932, 0.946]), against AUC 0.911 ([0.898, 0.925]) for the binary FL14 indicator (DeLong et al. 1988 test for the difference: $Z = -4.30$, $p = 2e - 05$). The continuous statistic is strictly more informative.

Price coefficients in standalone specifications point in the same direction (FL14 binary +0.0653, $p < 0.001$; continuous log(1+max tc) +0.0188, $p < 0.001$; Table 7, columns 1–2). When both are included additively (column 3), the FL14 coefficient flips sign (−0.0746, $p < 0.05$) while the continuous coefficient nearly doubles (+0.0349, $p < 0.001$). The flip is what the framework predicts: once the continuous primitive is in the specification, the binary indicator captures tail residuals against the continuous dosage rather than independent information about cover-bidder type. The construct's deployable rule and its discriminating instrument are not in tension—they are two views of the same latent posterior at different resolution.

The implication for deployment is clean. We adopt the binary FL14 rule as the headline operational implementation throughout the paper because it is administratively

interpretable (firms are either flagged or not, and the rule maps onto an existing oversight workflow). We adopt the continuous log(1 + tenders_count) as the discriminating instrument when ranking firms within the always-loser stratum, because it is the statistic the framework identifies as sufficient. The headline range +3.6%– + 7.7% in §7.1 is computed on the binary; the AUC headline 0.924 (95% CI [0.921, 0.926]) reported in the introduction and in §6 is the binary firm-level discrimination on the broader BEC-firm sample. Both are conservative readings of a continuous primitive that delivers strictly more information.

8.2. Modal Heterogeneity: Pregão Deeper than Convite

The construct fires with sharper signal-to-noise in pregão than in convite, on two independent dimensions of evidence. On the price-imprint side, the within-modal estimates reported in §7.1 place the binary frequent-loser specification at +9.59% ($p < 10^{-3}$) inside the pregão subsample versus +3.92% ($p = 0.037$) inside convite, a ratio of 2.45× in favor of pregão. On the discrimination side, the modal-by-modal AUC against the 193 adjudicated CADE-cobidder firms inside the always-loser stratum is 0.952 in pregão primary auctions versus 0.816 in convite primary auctions, with a bootstrap difference of -0.136 ($p \approx 0$). Both dimensions point in the same direction; neither is reducible to the other.

A naive reading would invoke convite's minimum-bidder rule (Lei 8.666/93, Art. 22, § 3) as the institutional driver of the construct—if statutorily mandated quorum-fillers are the mechanism, the construct should fire *more* cleanly in convite, not less. The data flatly reverse this prediction: the construct's price imprint and its discriminating power are both sharper where the minimum-bidder rule is absent. We do not read this reversal as positive evidence for any specific institutional theory of where cover bidding originates; we read it as scope information about where the construct's signal-to-noise is highest. Two competing readings remain observationally compatible at the firm level: pregão's open and electronic structure may expose loser-side participation footprints more visibly than convite's sealed-bid invitation procedure, or convite's smaller and more homogeneous bidder pools may compress the discriminating gap. We do not adjudicate between them,

because the design-based strategies (RDD, DiD) that would do so return null at the available bandwidths and the modal split is post-stratified rather than randomized.

The convite-modal AUC carries a width caveat the pregão-modal estimate does not: only 6 adjudicated CADE-cobidder firms fall in the within-convite stratum, so confidence intervals on the modal AUC difference are wider than the point estimate would suggest. We report the point estimate because it aligns in direction with the price-imprint evidence in §7.1, where the within-pregão coefficient strictly exceeds the within-convite coefficient on a much larger sample. For deployment, the operational implication is concrete: oversight bodies on platforms dominated by pregão-style auctions can expect stronger signal-to-noise from the construct than oversight bodies on convite-heavy registries. The cross-section of Brazilian state procurement platforms is heavily pregão, and the new federal procurement law (Lei 14.133/2021) consolidates the pregão-style electronic-auction format as the institutional default; the modal scope finding therefore strengthens, rather than weakens, the deployable case for the construct.

8.3. What the Data Do Not Support

Where §8.1 and §8.2 establish the construct's discrimination signature and its modal scope, this subsection refines its heterogeneity profile against the framework's sharper mechanism predictions. Two predictions—the cell-heterogeneity prediction across HHI and repeated-pair density, and the first-time-deployment prediction—do not adjudicate cleanly under disciplined estimation. We catalog them as scope information about where the construct's heterogeneity is most precisely identified.

The framework predicts that cover bidding should be most pronounced in low-concentration (low HHI) and low-repeated-pair-density environments, where rotation among cartelists is feasible without detection. The data place the largest positive conditional price gap exactly in the Low-HHI $\times$ Low-pairs cell, in the predicted direction. The signal carries one calibrating qualification: this cell is also the largest cell by item count, so the configuration is consistent with both the structural prediction and a generic compositional explanation, and our descriptive scope does not separate the two readings. The audit of the diagonally opposite High-HHI $\times$ High-pairs cell (Table 8) returns a

negative price coefficient (-0.062 to -0.119 depending on period and modality), with the negative sign concentrated in the 2009–2013 sub-period and in convite—period and modal compositional drivers rather than a clean refutation of the rotation prediction. Neither pattern adjudicates the cartel-rotation prediction; both are scope.

Table 8: Audit of the Negative High-HHI x High-Pairs Cell

Dimension	Group	FL coefficient	SE	p-value
modality	Convite	-0.062	0.024	0.009
modality	Pregao	-0.065	0.032	0.041
period	2014-2016	$+0.009$	0.033	0.796
period	2017-2019	-0.055	0.090	0.540
period	2009-2013	-0.119	0.025	< 0.001
direct_cade_item	0	-0.080	0.025	0.001

Notes: Each row re-estimates the within-item price association inside the problematic High-HHI x High-pairs cell. The purpose is not to rescue the mechanism, but to document where the sign flip lives. The full CSV additionally reports decompositions by buyer, item group, PBU size, and tender-value quartile.

The framework also predicts a discrete first-time-FL effect: when a firm crosses the threshold to frequent-loser status, its price imprint should sharpen. The estimate sits in a $+0.062$ to $+0.100$ range across three specifications, in the predicted direction throughout. The unconditional event-study estimate is $+0.100$ ($p = 0.019$, $N = 9{,}118$); under coarsened exact matching the estimate is $+0.085$ ($p = 0.076$, $N = 1{,}930$); under propensity-score matching it is $+0.062$ ($p = 0.312$, $N = 2{,}200$). Conditional matching compresses the magnitude substantially without inverting the direction. We adopt the propensity-score estimate as the conservative reference and report the full specification grid in Online Appendix C.

The construct's empirical content is the within-item conditional price gap of §7.1 and the procuring-unit-size oversight gradient of §7.2; the mechanism predictions of this subsection refine the construct's heterogeneity profile but are not its anchor. Read as a whole, the heterogeneity evidence places the construct in the framework's predicted direction along every dimension we examine, with the strength of the supporting evidence highest where the design provides the cleanest leverage and lowest where it does not.

9. Robustness

This section stress-tests the within-item conditional price gap of §7.1 and the procuring-unit-size oversight gradient of §7.2 against four distinct concerns. §9.1 addresses sensitivity to the design choices embedded in the construct (the IQR threshold multiplier, the clustering of standard errors, and the binary-vs-continuous specification of loss intensity). §9.2 reports identification audits (Cinelli and Hazlett 2020 robustness value, strict-overlap matching, leakage diagnostics, and a placebo on random firm assignment). §9.3 compares in-sample operational metrics to a temporal-holdout audit, the most consequential disclosure for deployment claims. §9.4 compares the construct's discrimination to bid-distribution detectors trained on bid microdata, treating the gap as informational complementarity rather than dominance.

9.1. Threshold and Specification Sensitivity

The headline price gap and its oversight gradient hold across reasonable variations in the design choices that define the construct. We document three: the IQR threshold multiplier, the clustering of standard errors, and the binary-versus-continuous specification of loss intensity.

The IQR multiplier governs the bright-line cutoff that separates frequent losers from the broader always-loser stratum. Our headline multiplier is 1.5, which yields a tender-count threshold of 13.5 and a flagged population of 2,735 firms. Table D.1 re-estimates the within-PBU specification under five multipliers ranging from 1.0× to 3.0× IQR. The price coefficient is positive and significant in every row, declining monotonically from +0.079 at 1.0× (threshold 10, 3,442 firms) to +0.050 at 3.0× (threshold 24, 1,456 firms). The decline is a feature, not a defect: as the cutoff tightens, the binary indicator filters fewer firms with high loss-intensity, and §8.1 established that the underlying primitive is the continuous $\log(1 + \text{tenders_count})$ statistic. The binary at 1.5× is the operational sweet spot between coverage and signal density; the construct neither evaporates at conservative thresholds nor inflates at permissive ones.

The clustering of standard errors does not change the point estimate—which is mechanical—but it does change inference. Table D.2 reports three regimes: item-level

clustering (baseline, SE 0.0215), procuring-unit-level clustering (SE 0.0144, tighter than baseline), and two-way Cameron et al. (2011) clustering on item and procuring unit jointly (SE 0.0240, looser than baseline). The coefficient is significant at $p < 0.01$ under all three regimes; the baseline item-level cluster is the conservative middle choice.

The binary-versus-continuous specification of loss intensity is discussed in detail in §8.1; we summarize the implication for robustness here. The continuous $\log(1+\text{tenders_count})$ score discriminates the same adjudicated CADE-cobidder population at AUC 0.939 (95% CI [0.932, 0.946]) versus 0.911 ([0.898, 0.925]) for the binary FL14 indicator (DeLong $Z = -4.30$, $p = 2e-05$). Standalone price coefficients in Table 7—FL14 binary +0.0653 ($p < 0.001$) and continuous +0.0188 ($p < 0.001$)—point in the same direction. The headline price gap is not an artifact of binary discretization.

9.2. Identification Audits

The construct's identification status is descriptive throughout the paper. This subsection collects four audits that calibrate the strength of the descriptive claim under specific identification concerns: omitted-variable sensitivity, common-support restriction, leakage, and a placebo on a non-cover-bidding channel.

The Cinelli and Hazlett (2020) robustness value reported in §7.1 is $RV_{q=1} = 17.5\%$: an unobserved confounder would need to explain at least 17.5% of the residual variation in both frequent-loser presence and log price to drive the coefficient to zero. The Oster (2019) bounds in Table C.1 agree directionally. The full-sample $\hat{\delta} = 261.64$ implies that unobservables would need to be more than two hundred times as important as the observables already captured by item, year, and procuring-unit fixed effects to overturn the result; the pregão subsample yields an even larger $\hat{\delta} = 634.00$. The convite subsample carries a smaller margin ($\hat{\delta} = 43.62$, $\beta^*(\delta = 1) = -0.1395$)— consistent with §8.2's finding that the construct's signal-to-noise is highest in pregão environments and that the convite estimate is the noisier modal slice. Both sensitivity tools agree: the broad-sample association is unlikely to be explained away by selection on observables of the kind already accounted for.

The strict-overlap matching exercise complements the sensitivity bounds with a

different identification angle. Where Cinelli and Oster ask how strong unobserved selection would need to be to overturn the result, strict-overlap matching restricts comparisons to the cells in which frequent-loser items and non-frequent-loser items genuinely overlap on observables. Table 9 reports four specifications. The baseline within-item-and-PBU coefficient is +0.064 (n = 1,653,658, identical to the headline). Restricting to overlap cells—requiring frequent-loser and non-frequent-loser items to share the same observed pre-treatment cell—returns −0.097 (n = 1,517,066). Trimming further to common-support under propensity-score restriction returns −0.307 (n = 400,687).

Table 9: Item-Level Scope Checks: Conditional Association Under Stricter Overlap

Specification	Coefficient	SE	p-value	N
baseline_fe	+0.064	0.021	0.003	1,653,658
overlap_cell_att	−0.097	0.015	< 0.001	1,517,066
overlap_ref_att	−0.097	0.015	< 0.001	1,434,636
ps_att_trimmed	−0.307	0.020	< 0.001	400,687

Notes: All rows remain descriptive. The overlap designs tighten comparability by forcing treated and untreated items into the same observed pre-treatment cells; the propensity-score row further trims to common support. None of these designs eliminates selection on unobservables, so the revised paper should call the resulting gap a conditional price association rather than a causal effect.

The reversal under strict overlap restricts comparisons to the items frequent losers *would* have entered with high probability, removing precisely the systematic selection differences between selected and unselected items that the broad-sample association captures (§7.1). Both signs describe the same population from different angles, neither carrying causal weight on its own; we report the four specifications jointly so the reader can locate the construct's descriptive claim within the appropriate scope.

The leakage audit in Table C.2 addresses a distinct concern: that the in-sample item-level AUC of 0.995 (reference row) is partially tautological, because cobidders are by construction always-loser firms with elevated participation counts. Audit 1 holds the score fixed and swaps the label to direct-CADE defendants, returning 0.506—random discrimination, confirming the structural scope of the construct (cover bidders, not

ringleaders). Audit 2 breaks the construction by withholding cobidder firms from the score-construction step within five cross-validation folds and re-evaluating on items containing held-out cobidders: AUC 0.891 (95% CI $[0.887, 0.894]$). Audit 3 breaks the construction temporally: train 2009–2016, test 2017–2019, returning AUC 0.864 (95% CI $[0.859, 0.870]$). The screen retains discriminating power above 0.85 under both audits. The drop of 0.10–0.13 is the pure-leakage component, and we adopt the temporal-holdout AUC of 0.864 as the conservative discriminating reference for the operational-metrics audit in §9.3.

A placebo specification using the supply of sub-threshold always-losers (firms with persistent zero wins but below the IQR cutoff) as an instrument completes the audit set. The first-stage F-statistic is 33, comfortably above the Stock and Yogo (2005) weak-instrument threshold, but the 2SLS coefficient is 5.24 (Table 10, below)—an implausible magnitude that signals the placebo instrument operates through channels other than cover bidding. The construct's price imprint draws on the above-threshold cover-bidder population we identify, not on generic always-loser supply.

9.3. Operational Metrics: In-Sample vs Temporal Holdout

The deployment claim rests on operational metrics computed under the temporal-holdout regime, not under in-sample evaluation. The construct's discriminating performance under temporal holdout is firm-level AUC 0.864 (95% CI $[0.859, 0.870]$) against the 193 adjudicated CADE-cobidder labels, with the score constructed using only 2009–2016 participation counts and evaluated on items in 2017–2019. This is the operationally honest projection for a screening tool that is built on past participation data and deployed against a future case-load. We report it as the operational reference throughout the paper.

Table E.1 reports precision, recall, and lift at five top-k cuts under both in-sample and temporal-holdout evaluation. In the deployment-relevant region, top-100 has precision 0.070 (lift 6.1×), top-500 has precision 0.070 (lift 6.1×), and top-1,000 has precision 0.066 (lift 5.8×). At top-500, the screen flags 35 adjudicated cobidder firms out of 193 in the pool (recall 18%); at top-1,000 the recall rises to 34%. The construct's signal

Table 10: Placebo IV: Sub-Threshold Always-Losers as Instrument

	First Stage	2SLS
	FL presence	Log Price
Sub-threshold supply (LOO)	0.000040***	
	(0.000007)	
F-statistic	33.0	
FL presence (instrumented)		5.2445***
		(1.3200)

Notes: Placebo instrument uses supply of sub-threshold always-losers (below the IQR cutoff). The sub-threshold first stage ($F = 33$) exceeds the Stock and Yogo (2005) 10% maximal IV size critical value of 16.38 for a single instrument. The implausibly large 2SLS coefficient (5.24) is therefore not attributable to weak-instrument bias. Rather, it reflects the absence of a genuine cover-bidding mechanism: sub-threshold always-losers do not drive procurement price anomalies, confirming that the main instrument's price effect operates through the cover-bidding channel rather than through correlated demand shocks. SE clustered at item level.

sustains through the deployable region, with lifts in the 5.8× to 6.6× range across cuts. The in-sample column over-states precision by approximately 50% at top-500 (0.132 in-sample versus 0.070 holdout, retention 53%). The inflation arises because roughly 47% of the top-500 in-sample ranking comes from 2017–2019 participation, after CADE adjudications were already underway for some of the cartels, so the in-sample score is partly retrospective rather than purely prospective. We disclose the gap explicitly and operate on the temporal-holdout column.

Three operational implications follow. First, the construct is deployable: discriminating AUC 0.864 and lifts of 5.8× to 6.6× in the top-100 to top-1,000 region are operationally meaningful for a screening tool that runs on contract-award records alone, at near-zero marginal data cost. Second, the temporal holdout is the honest reference: any operational claim in this paper that requires a precision or lift number is computed on the holdout column, and the in-sample column is reported for comparison and transparency. Third, periodic rescoring matters: an oversight body running the screen in production should refresh the score window as new years of contract-award data arrive, both to dampen drift and to keep the prospective component dominant in the deployment ranking.

9.4. Horse Race Against Bid-Distribution Detectors

The construct's competitive position vis-à-vis bid-distribution detectors that require bid microdata is the final calibration. The relevant comparison is at fixed *data cost*, not at fixed sample, because the two pipelines operate on different data envelopes: the construct uses award-record fields only (winner identity, participant identity, item identifier), while the bid-distribution pipeline requires per-bid offer values.

Table 11 reports the comparison on the full always-loser pool (N = 16,779 firms, 193 adjudicated cobidder labels). The construct discriminates at AUC 0.903 (95% CI $[0.884, 0.923]$) under the binary FL14 indicator and 0.884 (95% CI $[0.860, 0.908]$) under the continuous $\log(1 + \text{tenders_count})$ score. The full Imhof–Wallimann pipeline—five within-tender bid-distribution features (coefficient of variation, skewness, kurtosis, spread, second-low ratio) (Imhof, 2019; Wallimann et al., 2023)—reaches AUC 0.888 (95% CI $[0.865, 0.911]$) on the same target. The two pipelines reach comparable discrimination

on different data envelopes. We do not claim that the construct outperforms the bid-distribution pipeline; we claim that it reaches comparable accuracy on a substantially thinner data envelope.

Table 11: Full Imhof–Wallimann Pipeline vs Loser-Side Concentration: Discrimination at Different Data Costs

Model	Features required	AUC	95% CI
Participation-only (award-record level)			
FL flag (binary)	participation count > 14, win flag	0.903	$[0.884, 0.923]$
log(1+tenders_count)	participation count, win flag	0.884	$[0.860, 0.908]$
Bid-distribution (per-bid microdata)			
Imhof CV only (v13 strawman)	per-bid offer values	0.585	$[0.553, 0.616]$
Imhof full pipeline	5 within-tender features (CV, skew, kurt, spread, second-low)	0.888	$[0.865, 0.911]$
Combined			
FL flag + Imhof full	participation count + 5 bid features	0.955	$[0.943, 0.967]$
log(1+tenders_count) + Imhof full	participation count + 5 bid features	0.962	$[0.954, 0.969]$

Notes: 5-fold cross-validated AUC against 193 CADE-cobidder labels within the always-loser pool ($N = 16{,}779$). Random forests with 500 trees, otherwise default tuning. Bid-level features are aggregated to the firm level by averaging the per-tender within-tender statistic across each firm's tenders. The two top blocks compare alternative single-construct screens operating on different data envelopes; the bottom block reports the combined classifiers. The combined model gains 5–7 AUC points over either single construct, indicating non-redundant signal in the participation-only and bid-distribution information sources. The participation-only screen achieves discrimination comparable to the full Imhof–Wallimann pipeline at substantially lower data cost, which is the operational claim of the paper. We do not claim the participation-only screen *outperforms* the bid-distribution pipeline; the relevant comparison is at fixed data cost, where the two pipelines reach similar accuracy and the combination dominates.

The combined specifications in the bottom panel of Table 11 address the integration question. FL14 combined with the full Imhof pipeline reaches AUC 0.955 (95% CI $[0.943, 0.967]$); continuous loss intensity combined with the full Imhof pipeline reaches AUC 0.962 (95% CI $[0.954, 0.969]$). Both combined classifiers gain 5–7 AUC points over either single construct, indicating non-redundant signal in the participation-only and bid-distribution information sources. Restricting attention to the same-sample audit in Table E.2, where all detectors evaluate the identical subset of firms with bid microdata available, the construct adds +0.035 AUC over the Imhof full pipeline alone (DeLong $p = 0.014$), and the combined model adds +0.096–+0.098 AUC over Imhof full alone (DeLong $p < 0.001$).

The construct's competitive position is therefore informational complementarity, not dominance. It reaches comparable discrimination at lower data cost when bid microdata are unavailable, and it adds non-redundant signal when bid microdata are available. The

administrative pathway in §11 integrates the construct with the bid-distribution stage rather than substituting for it: the **Screen** stage runs on award records, the **Forensic** stage deploys bid-distribution methods, and the gain from running both is identifiable in the data.

10. Limitations

The empirical content of the paper has a deliberately narrow scope. This section consolidates the limitations along three dimensions: what identification can deliver in §10.1, what the construct flags and what it does not in §10.2, and how the empirical claim travels beyond São Paulo's electronic procurement platform in §10.3.

10.1. Identification scope

Identification is descriptive throughout the paper. Two design-based strategies that would extract a causal estimate from the data return null at the available bandwidths: a sharp regression discontinuity at the convite/pregão statutory caps, and a difference-in-differences exploiting the 2018 cap raise. The first is gated by the absence of a McCrary-test-passing density around the institutional thresholds in the BEC sample; the second is gated by the absence of a clean pre-trend in the relevant outcome series. We report the failures openly and treat the empirical content as a descriptive association rather than as a causal estimate.

The strict-overlap matching exercise developed in §7.1 and audited in §9.2 is the most informative single observation about the limits of the descriptive claim. The within-PBU coefficient of 6.4% on the broad sample reverses to −9.7% under matching restricted to overlap cells and to −30.7% under propensity-score trimming (Table 9). Neither the broad-sample association nor the strict-overlap inversion carries causal weight on its own. Both are conditional descriptions of the same population from different angles: the broad-sample gap captures the systematic differences between items frequent losers select into and items they do not, and the strict-overlap inversion removes those differences by construction. The sensitivity bounds (Cinelli and Hazlett (2020) robustness value 17.5% and Oster (2019)'s $\hat{\delta} = 261.64$ on the full sample) discipline the broad-sample

association against selection on observables of the kind already captured by item, year, and procuring-unit fixed effects, but they do not extend to the selection on unobservables that strict-overlap matching surfaces.

What this implies for the paper's claims is concrete. The $+3.6\% - +7.7\%$ headline range is a conditional descriptive association across estimators, not an estimate of cover-bidding's causal effect on prices. The 12.6× procuring-unit-size oversight gradient is consistent with the framework's comparative static $\partial m^*/\partial \theta_k < 0$, but the consistency is observational; the gradient does not identify the causal channel through which oversight infrastructure moderates the participation footprint. For the construct's intended use—prioritizing oversight case-load on environments where the participation footprint is institutionally structured—this descriptive scope is sufficient. A screen does not need to identify causal effects to allocate scarce auditing resources; it needs to flag environments at higher prior risk than random selection would, and to do so on data the oversight body already holds.

10.2. Construct scope

The construct flags loser-side participation patterns, not cartel members. Inside the always-loser stratum where the construct operates, it discriminates the 193 adjudicated CADE-cobidder firms at firm-level AUC 0.924 (95% CI [0.921, 0.926]) in sample and at 0.864 (95% CI $[0.859, 0.870]$) under temporal holdout (§9.3). Against the 47 direct CADE defendants in the broader BEC firm universe, the construct discriminates at AUC 0.49, essentially random. The asymmetry is by design, not a failure: direct defendants in adjudicated cases are typically frequent winners, while the construct concentrates on participation footprints characterized by persistent zero wins. A deploying oversight body that wants to detect cartel ringleaders directly will need a different tool; the construct flags the loser side of the cartel population, and the empirical AUC differential against the two populations is what that scope looks like in the data.

The participation footprint the construct flags is observationally compatible with at least three non-competitive readings, and our evidence does not adjudicate among them at the firm level. The working reading we adopt throughout the paper is cover bidding

under cartel rotation, derived from the framework in Online Appendix A. Two competing readings remain: a *capacity-transfer* reading, in which firms maintain bidding presence to signal capacity availability for subcontracting or for affiliate consortia; and a *market-intelligence* reading, in which firms participate to observe prices, identify procurement schedules, or build operational priors on competitor behavior. These readings are not mutually exclusive, and the same firm can have multiple non-competitive motives. The price imprint of §7.1 and the oversight gradient of §7.2 are observationally consistent with all three. We adopt the cover-bidding reading because the framework derives from it and because the strongest heterogeneity dimensions—the modal asymmetry between pregão and convite (§8.2) and the procuring-unit-size oversight gradient (§7.2)—match its comparative statics most directly. The adoption is a working interpretation, not an identification claim.

Two further mechanism predictions that the framework generates remain undiscriminated by the data. The cell-heterogeneity prediction across HHI and repeated-pair density places the largest positive price gap in the Low-HHI $\times$ Low-pairs cell, in the predicted direction (§8.3); but this cell is also the largest cell by item count, leaving the configuration observationally compatible with both the structural prediction and a generic compositional explanation. The first-time-FL prediction is in the predicted direction throughout (+0.062 to +0.100 across specifications), with conditional matching compressing the magnitude substantially without inverting it (§8.3). Neither mechanism prediction is refuted; neither is cleanly confirmed.

The construct's behavior under deliberate adversarial adaptation is bounded but not invariant. We simulate four strategic adaptations holding the construct fixed (Online Appendix G, Table F.1): periodic rotation of cover bidders, occasional wins to evade the bright-line filter, CNPJ splitting, and threshold-aware participation capping. The construct is resilient to rotation and occasional wins (AUC essentially unchanged), degrades modestly under CNPJ splitting (-3.7 points), and is substantially vulnerable to threshold capping (-12.7 points). Combining threshold capping with rotation drives AUC from 0.928 to 0.632. We do not claim the construct is adaptation-proof. The threshold rule is the most exposed margin, and the construct's operational lifetime in any

deployment requires periodic recalibration and integration with bid-distribution methods where those data are available. The construct flags a participation footprint with multiple non-competitive readings, the supporting evidence places it in the framework's predicted direction along every dimension we examine, and the strength of the supporting evidence is highest where the design provides the cleanest leverage.

10.3. External validity and portability

The empirical evidence in this paper comes from a single jurisdiction and a single time window: São Paulo's Bolsa Eletrônica de Compras, 2009–2019. The construct's portability beyond this envelope rests on three structural features of the procurement environment, all explicit in BEC and identifiable in any candidate jurisdiction before deployment: a commodity-heavy item composition that supports within-product-code price comparisons; an electronic platform with publicly auditable participant lists, so that participation counts can be computed without privileged data access; and an institutional asymmetry between the winner and loser sides of the bidding pool that makes persistent zero wins informative about strategic non-competitive participation rather than about randomized matching outcomes. These are diagnostic conditions, not optional ones.

Settings that this paper does not test directly are several. The federal Brazilian procurement registry (ComprasNet) and other state-level registries (Banco do Brasil, individual UF platforms) preserve the same award-record envelope as BEC and are accessible through the same legal regime; replication is technically straightforward and substantively pending. Non-commodity settings (services, infrastructure, public works) are within the framework's logical scope but outside the empirical scope of the present work, and the within-item identifying variation that underwrites the headline price gap is not equally available in those settings. Time periods outside the sample window are likewise untested: the BEC sample begins after the platform's operational stabilization and ends before the consolidation of the new federal procurement law, and the construct's stability across regulatory regimes is an open empirical question.

The new federal procurement law (Lei 14.133/2021) consolidates the pregão-style electronic-auction format as the institutional default for Brazilian federal and inter-state

procurement. The modal-asymmetry result of §8.2, that the construct's signal-to-noise is highest in pregão, means that the policy direction is favorable to the construct's portability, not adversarial. The construct's three structural prerequisites are increasingly the structural defaults of Brazilian public procurement, and the deployment case for state and federal oversight bodies operating on the new institutional baseline is correspondingly stronger. Portability remains contingent, not free; but the contingencies the construct requires are recognizable, observable, and, in the institutional direction the country is moving, increasingly satisfied.

11. Conclusion

The participation-based screen we build extracts usable signal for procurement triage from the data envelope every procurement system already maintains. Frequent-loser presence on contract-award records associates with a $+3.6\% - +7.7\%$ higher conditional log negotiated unit price across estimators, and discriminates cartel-adjacent populations at rates well above random matching: AUC 0.748 on a conservative contemporaneous benchmark restricted to four pre-2020 cases, and 0.864 on temporal holdout against a more permissive prospective benchmark. These numbers describe the screen's operational footprint, not cartel membership; the validation object is adjacency to documented collusion, and the construct discriminates direct CADE defendants in the broader BEC firm universe at AUC 0.49, by design.

Two interpretive caveats anchor the reading. The conditional price association reverses sign under strict-overlap matching (-9.7% to -30.7%): the broad-sample estimate combines a within-cell association with the systematic differences between items frequent losers select into and items they do not, and the screen's signal is partly about that selection. The price imprint shrinks 12.6× across buyer-size quartiles, consistent with thinner institutional capacity moderating the imprint at smaller buyers, but buyer size proxies for several correlated features and we do not identify oversight as the operative channel.

The screen is designed for triage, computable on award records alone, and integrates with bid-distribution methods where bid microdata are available (Imhof, 2019; Walli-

mann et al., 2023). Adversarial adaptation degrades performance predictably (Online Appendix G), replication beyond São Paulo remains an open empirical question, and the contribution stops short of cartel detection by design.

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Author Contributions (CRediT)

Darcio Genicolo-Martins: conceptualization, data curation, formal analysis, investigation, methodology, software, validation, visualization, writing—original draft, writing—review and editing. **Paulo Furquim de Azevedo:** conceptualization, methodology, supervision, writing—review and editing.

Data and Code Availability

The CADE adjudication records used as ground truth are public and accessible through the *Conselho Administrativo de Defesa Econômica* portal (`https://www.gov.br/cade`). The São Paulo Bolsa Eletrônica de Compras (BEC) data are administrative records maintained by the State of São Paulo Treasury and were obtained through institutional channels. The replication archive containing the R, Python, and LaTeX source files, the construction scripts for all derived datasets, and the git history that produced the analysis is available upon publication via the journal's data-availability portal. The archive includes: (i) the canonical pipeline (`00_master.R` plus the build, clean, and analysis scripts) that regenerates all tables and figures from the four primary parquet files (`firm_loss_stats.parquet`, `firm_tender_map.parquet`,

`FREQ_PARTICIP_rebuilt.parquet`, `LOSERS_rebuilt.parquet`); (ii) the institutional-identification scripts producing Table B.5; (iii) the cross-sector replication script producing Table B.4; and (iv) the inverse-CNAE mapping diagnostic discussed in Section 10.3.

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Declarations of Interest

None.

Reproducibility Note

The numerical claims in this paper come from a single pipeline run on the canonical 2009–2019 BEC parquet files plus the CADE crossmatch CSV. The pipeline applies a strict 14-character CNPJ zero-padding convention before every merge: this padding is required to recover the full 193 FL–CADE co-bidder set, and alternative conventions yield the 98-firm subset that appears in earlier draft tables. We flag the convention here so future replications reproduce either count deterministically. The headline AUC of 0.924 corresponds to point estimate 0.9389 on 193 positives among 16,843always-losers; the 95% confidence interval [0.932, 0.946] uses DeLong's method, with stratified percentile and BCa bootstrap intervals ($B = 5,000$ and $2,000$, respectively) agreeing to three decimal places. The cross-sector replication (§10.3) runs on the same canonical sample with the BEC item-code 2-digit prefix as the sector partition; the inverse CNAE mapping (§10.3 footnote) uses the `Firms_final.parquet` CNAE codes recovered for 34 of the 47 BEC-active CADE-defendants. The full sequence of commits that produced the final tables and figures lives in the replication archive's git log.

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